

Homework Classical Physics, Fall 2023

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Problem Set 03.2 (L05 & L06)

Due on Thursday, 28. September 2023, at 09:30 (in the morning!)

(= Deadline for submitting a pdf file with solution)

Mechanics - Force and acceleration (4 points)

A load of $M = 850 \text{ kg}$ is moved by a crane. Determine the force in the rope of the crane when

- (a) the load is moved upwards with a constant increase of velocity of 1.6 m/s per second,
- (b) the load is moved upwards with constant velocity,
- (c) the load is lowered with a decrease in velocity of 2.1 m/s per second.
- (d) What is the theoretical maximum acceleration with which you can pull at the rope with the load attached assuming it is specified to rupture at loads above 1500 kg ?